

Differential Equations Lesson 09

Question 1:

The second-order linear homogeneous differential equation: $x^2 \cdot y'' - 9x \cdot y' + 25y = 0$, where $x > 0$.

Given solution: $y_1(x) = x^5$.

Find a second solution of the differential equation by the method of reduction of order.

The differential equation $y'' - \frac{9x}{x^2} \cdot y' + \frac{25}{x^2} y = 0$ is of the form $y'' + P(x) \cdot y' + Q(x) \cdot y = 0$.

Let $P(x) = -\frac{9x}{x^2} = \frac{-9}{x}$

a) $e^{-\int P(x) \cdot dx} = \underline{\hspace{2cm}} ?$

b) $\int \frac{e^{-\int P(x) \cdot dx}}{[y_1(x)]^2} \cdot dx = \underline{\hspace{2cm}} ?$

c) The second solution is $y_2(x) = y_1(x) \cdot \int \frac{e^{-\int P(x) \cdot dx}}{[y_1(x)]^2} \cdot dx = \underline{\hspace{2cm}} ?$

d) Are the functions $y_1(x)$ and $y_2(x)$ linearly independent?

e) The general solution of the differential equation $x^2 \cdot y'' - 9x \cdot y' + 25y = 0$ is $\underline{\hspace{2cm}} ?$.

Hint: general solution should be of the form: $y(x) = C_1 \cdot y_1(x) + C_2 \cdot y_2(x)$

Question 2:

The second-order linear homogeneous differential equation: $y'' + 5y' - 6y = 0$.

Given solution: $y_1(x) = e^x$.

Find a second solution of the differential equation by the method of reduction of order.

Note: $y'' + 5y' - 6y = 0 \Rightarrow y'' + 5y' - 6y = 0$.

The differential equation $y'' + 5y' - 6y = 0$ is of the form $y'' + P(x) \cdot y' + Q(x) \cdot y = 0$.

Let $P(x) = 5$.

a) $e^{-\int P(x) \cdot dx} = \underline{\hspace{2cm} ? \hspace{2cm}}$

b) $\int \frac{e^{-\int P(x) \cdot dx}}{[y_1(x)]^2} \cdot dx = \underline{\hspace{2cm} ? \hspace{2cm}} \square$

c) The second solution is $y_2(x) = y_1(x) \cdot \int \frac{e^{-\int P(x) \cdot dx}}{[y_1(x)]^2} \cdot dx = \underline{\hspace{2cm} ? \hspace{2cm}}$

d) Are the functions $y_1(x)$ and $y_2(x)$ linearly independent?

e) The general solution of the differential equation $y'' + 5y' - 6y = 0$ is ____?

Hint: general solution should be of the form: $y(x) = C_1 \cdot y_1(x) + C_2 \cdot y_2(x)$

□

Question 3:

The second-order linear homogeneous differential equation: $y'' + 16y = 0$.

Given solution: $y_1(x) = \cos(4x)$.

Find a second solution of the differential equation by the method of reduction of order.

Note: $y'' + 16y = 0 \Rightarrow y'' + 0y' + 16y = 0$.

The differential equation $y'' + 0y' + 16y = 0$ is of the form $y'' + P(x) \cdot y' + Q(x) \cdot y = 0$.

Let $P(x) = 0$.

a) $e^{-\int P(x) \cdot dx} = \underline{\hspace{2cm} ? \hspace{2cm}}$

b) $\int \frac{e^{-\int P(x) \cdot dx}}{[y_1(x)]^2} \cdot dx = \underline{\hspace{2cm} ? \hspace{2cm}}$

c) The second solution is $y_2(x) = y_1(x) \cdot \int \frac{e^{-\int P(x) \cdot dx}}{[y_1(x)]^2} \cdot dx = \underline{\hspace{2cm}} ?$

d) Are the functions $y_1(x)$ and $y_2(x)$ linearly independent?

e) The general solution of the differential equation $y'' + 16y = 0$ is $\underline{\hspace{2cm}} ? \underline{\hspace{2cm}}$.

Hint: general solution should be of the form: $y(x) = C_1 \cdot y_1(x) + C_2 \cdot y_2(x)$

Question 4:

The second-order linear homogeneous differential equation: $y'' + 5y' + 10.25y = 0$.

Given solution: $y_1(x) = e^{-2.5x} \cos(2x)$.

Find a second solution of the differential equation by the method of reduction of order.

The differential equation $y'' + 5y' + 10.25y = 0$ is of the form $y'' + P(x) \cdot y' + Q(x) \cdot y = 0$.

Let $P(x) = 5$.

a) $e^{-\int P(x) \cdot dx} = \underline{\hspace{2cm}} ?$

b) $\int \frac{e^{-\int P(x) \cdot dx}}{[y_1(x)]^2} \cdot dx = \underline{\hspace{2cm}} ?$

c) The second solution is $y_2(x) = y_1(x) \cdot \int \frac{e^{-\int P(x) \cdot dx}}{[y_1(x)]^2} \cdot dx = \underline{\hspace{2cm}}?$

d) Are the functions $y_1(x)$ and $y_2(x)$ linearly independent?

e) The general solution of the differential equation $y'' + 5y' + 10.25y = 0$ is ?.

Hint: general solution should be of the form: $y(x) = C_1 \cdot y_1(x) + C_2 \cdot y_2(x)$

